

Product Rule For Counting Worksheet

Fundamentals:

1 Janey is choosing a meal which consists of a starter and a main.

There are 3 options for a starter, and 4 options for a main.

How many different meals are possible?

Answer _____

2 Nelson is choosing an outfit, which will consist of a shirt, shoes, and trousers.

He has 2 options for his shirt, 4 options for shoes. He works out that there are 24 different possible outfits.

How many options for trousers does Nelson have?

Answer _____

3 Martin wants to create a 3-digit pin for his phone. He doesn't want any digits to be repeated.

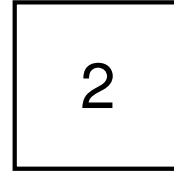
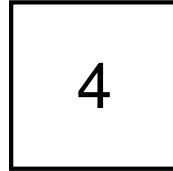
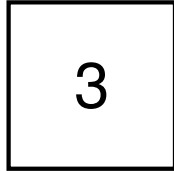
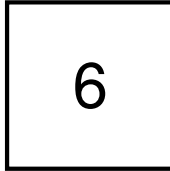
Using digits from 0 to 9, how many different ways can he do this?

Answer _____

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Intermediate:

1 Ralph wants to organise the following 4 tiles into a 4 digit number.



How many different ways can he do this and make an **odd** number?

Answer _____

How many different ways can he do this and make an **even** number?

Answer _____

2 Sherri chooses a 3 digit **even** number. The first digit of her number is odd.

How many different numbers satisfy her criteria?

Answer _____

3 Terri chooses a 4 digit odd number.

Her number starts with a factor of 9 and ends with a multiple of 3.

How many different numbers satisfy her criteria?

Answer _____

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Challenge:

1 How many 4 digit numbers are there which do not repeat any digits?

Answer _____

2 How many 5 digit numbers are there which are not multiples of 5?

Answer _____

3 Sashi writes down a 5 digit number.

The first digit is prime.

The fourth digit is a factors of 36.

The last digit is a repeat of one of the other 4 digits.

How many different 4 digit numbers could Sashi have written down?

Answer _____